# RSpec. Review

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# Сергей Удалов

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# Как избавить себя и команду от боли при работе с RSpec?


## Сергей Удалов

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в Samokat Tech

ruby
russia'23

TODO: Зачем это доклад?

# Рассмотрим

dry-rb gitlab hanami pg rom-rb
rspec ruby-concurrency vcr

# План

1. Конфигурация
2. Контекст
3. Тестовые классы
4. Expectation
5. Faker
6. Helpers
7. Выразительность
8. Acceptance
9. Библиотеки

# Конфигурация

# helper 1/2

- spec_helper
- fast_spec_helper
- rake_helper
- rubocop_spec_helper

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec

# helper 2/2

```
RSpec.configure do |config|
  config.define_derived_metadata(file_path: %r{spec/rubocop}) do |metadata|
    metadata[:type] = :rubocop
  end

  config.include RuboCop::RSpec::ExpectOffense, type: :rubocop
  config.include_context 'config', type: :rubocop
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/rubocop_spec_helper.rb

```ruby
it 'should tell about pipeline mode', :postgresql_14 do
  @conn.enter_pipeline_mode
  expect(@conn.inspect).to match(/ pipeline_status=PQ_PIPELINE_ON/)
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/connection_spec.rb

```ruby
# TODO: each spec file should require its dependencies.
# Maybe we'll get there one day
require "rom"
require "rom/sql"
require "rom/memory"
require "rom/repository"
require "rom/changeset"
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/spec_helper.rb

```
config.define_derived_metadata file_path: %r{/suite/} do |metadata|
  metadata[:group] = metadata[:file_path]
                     .split('/')
                     .then { |parts| parts[parts.index('suite') + 1] }
                     .to_sym
end

%i[rom compat].each do |group|
  config.when_first_matching_example_defined group: group do
    require_relative "support/#{group}"
  end
end
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/spec_helper.rb

# Helper 1/3

```
RSpec.describe ROM::SQL::Associations::ManyToMany, helpers: true do
```

https://github.com/rom-rb/rom-sql/blob/beb115/spec/integration/associations/many_to_many_spec.rb

# Helper 2/3

```
config.include(Helpers, helpers: true)
```

https://github.com/rom-rb/rom-sql/blob/beb115/spec/spec_helper.rb

# Helper 3/3

```
Dir[root.join("shared/**/*.rb")].sort.each { |f| require f }
Dir[root.join("support/**/*.rb")].sort.each { |f| require f }
```

https://github.com/rom-rb/rom-sql/blob/beb115/spec/spec_helper.rb

# Sanboxed

```
RSpec.configure do |c|
  c.around do |ex|
    RSpec::Core::Sandbox.sandboxed do |config|
      # If there is an example-within-an-example,
      # we want to make sure the inner example
      # does not get a reference to the outer example
      # (the real spec) if it calls
      # something like `pending`
      config.before(:context) { RSpec.current_example = nil }

      config.color_mode = :off

      orig_load_path = $LOAD_PATH.dup
      ex.run
      $LOAD_PATH.replace(orig_load_path)
    end
  end
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/support/sandboxing.rb

# Контекст

```
▾ fixtures/
  ▾ app/persistence/
    ▸ commands/
    ▸ mappers/
    ▸ my_commands/
    ▸ my_mappers/
    ▸ my_relations/
    ▸ relations/
  ▾ auto_loading/
    ▾ app/
      ▸ commands/users/
      ▸ mappers/
      ▸ relations/
    ▸ persistence/
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/fixtures

```ruby
RSpec.shared_context 'changeset / database' do
  include_context 'changeset / database setup'

  before do
    %i[tags tasks books posts_labels posts users labels
       reactions messages].each { |table| conn.drop_table?(table) }

    conn.create_table :users do
      primary_key :id
      column :name, String
    end

    conn.create_table :books do
      primary_key :id
      foreign_key :author_id, :users, on_delete: :cascade
      column :title, String
      column :created_at, Time
      column :updated_at, Time
    end
  end
end
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/shared/rom/changeset/database.rb

# attr_reader

```
context 'when used in combination with the BisectDRbFormatter', :slow do
  include FormatterSupport

  attr_reader :server

  around do |ex|
    Bisect::Server.run do |the_server|
      @server = the_server
      ex.run
    end
  end

  def run_formatter_specs
    RSpec.configuration.drb_port = server.drb_port
    run_rspec_with_formatter('bisect-drb')
  end

  it 'receives suite results' do
    results = server.capture_run_results(['spec/rspec/core/resources/formatter_specs.rb']) do
    end
  end
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/rspec/core/bisect/server_spec.rb

# Перменные вне блоков

```
RSpec.describe(Dry::Monads) do
  let(:m) { described_class }
  list = Dry::Monads::List

  it "builds a list with List[]" do
    expect(instance.make_list).to eql(list[1, 2, 3])
  end
end
```

https://github.com/dry-rb/dry-monads/blob/704c1b/spec/integration/monads_spec.rb

# `as_null_object` 1/2

```
allow(config).to receive(:logger).and_return(double.as_null_object)
```

https://github.com/vcr/vcr/blob/8f5636/spec/lib/vcr/cassette/http_interaction_list_spec.rb

# `as_null_object` 2/2

```
logger = double.as_null_object
expect(logger.foo.bar).to be_nil?
```

# Инкапсуляция

```
describe 'clean_outdated_http_interactions' do
  before(:each) do
    subject.instance_variable_set(:@clean_outdated_http_interactions, true)
    subject.instance_variable_set(:@previously_recorded_interactions,
                                  subject.instance_variable_get(:@new_recorded_interactions))
    subject.instance_variable_set(:@new_recorded_interactions, [])
  end
end
```

https://github.com/vcr/vcr/blob/8f5636/spec/lib/vcr/cassette_spec.rb

# Тестовые классы

```ruby
it 'raises no error when all monads are loaded' do
  expect do
    class Test::MyClass
      include Dry::Monads
    end
  end.not_to raise_error

  expect(Test::MyClass.constants).to include(:Success)
  expect(Test::MyClass.constants).to include(:Failure)
end
```

https://github.com/dry-rb/dry-monads/blob/704c1b/spec/integration/all_monads_inclusion_spec.rb

```ruby
before do
  configuration.relation(:users)
  configuration.relation(:tasks)

  class Test::CreateUser < ROM::Commands::Create[:memory]
    config.component.id = :create
    config.component.namespace = :users
    config.component.relation = :users
    config.result = :one
  end
end
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/suite/rom/commands/create_spec.rb

```ruby
subject do
  Class.new do
    def zero
      nil
    end

    def three(_a, _b, _c)
      nil
    end

    def two_plus_two(_a, _b, _c = nil, _d = nil)
      nil
    end

    def many(*_args)
      nil
    end
  end.new
end
```

https://github.com/ruby-concurrency/concurrent-ruby/blob/1982b9/spec/concurrent/async_spec.rb

```ruby
it "shouldn't accept invalid return from fit_to_copy_get" do
  tm = Class.new(PG::TypeMapInRuby) do
    def fit_to_copy_get
      :invalid
    end
  end.new.freeze

  ce = PG::TextDecoder::CopyRow.new(type_map: tm).freeze
  expect { ce.decode("5\t6\n") }.to raise_error(TypeError, /kind of Integer/)
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/type_map_in_ruby_spec.rb

```ruby
module Test
  def self.remove_constants
    constants.each(&method(:remove_const))
  end
end

config.after do
  gateway.disconnect if respond_to?(:gateway)
    && gateway.respond_to?(:disconnect)
  Test.remove_constants
end
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/spec_helper.rb

# Expectation

```
it "returns a new instance" do
  expect(none).to eql(None())
  expect(none.trace).to include("spec/integration/\
    maybe_spec.rb:9:in `block")
end
```

https://github.com/dry-rb/dry-monads/blob/704c1b/spec/integration/maybe_spec.rb

```ruby
let(:logger_stream) { StringIO.new }

def configure_logger
  Hanami.app.config.logger.stream = logger_stream
end

def logs
  @logs ||= (logger_stream.rewind and logger_stream.read)
end

expect(logs).to match %r{GET 404 \d+(µs|ms) 127.0.0.1 /}
```

https://github.com/hanami/hanami/blob/675b44/spec/integration/logging/exception_logging_spec.rb

```ruby
describe 'sessions' do
  specify { expect(config.sessions).not_to be_enabled }
end
```

https://github.com/hanami/hanami/blob/a2bdb7/spec/unit/hanami/config/actions/default_values_spec.rb

# Несколько проверок 1/3

```
let!(:textdec_int) { PG::TextDecoder::Integer.new(name: 'INT4', oid: 23).freeze }
let!(:textdec_float) { PG::TextDecoder::Float.new(name: 'FLOAT8', oid: 701).freeze }
let!(:textdec_string) { PG::TextDecoder::String.new(name: 'TEXT', oid: 25).freeze }
let!(:textdec_bytea) { PG::TextDecoder::Bytea.new(name: 'BYTEA', oid: 17).freeze }
let!(:binarydec_float) { PG::BinaryDecoder::Float.new(name: 'FLOAT8', oid: 701, format: 1).freeze }
let!(:tm) do
  tm = PG::TypeMapByOid.new
  tm.add_coder textdec_int
  tm.add_coder textdec_float
  tm.add_coder binarydec_float
  tm.add_coder pass_through_type
  tm.freeze
end
it "should retrieve it's conversions" do
  expect(tm.coders).to eq([
                            textdec_int,
                            textdec_float,
                            pass_through_type,
                            binarydec_float
                          ])
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/type_map_by_oid_spec.rb

# Несколько проверок 2/3

```
it 'should check format when deleting coders' do
  expect { tm_writable.rm_coder(2, 123) }.to raise_error(ArgumentError)
  expect { tm_writable.rm_coder(-1, 123) }.to raise_error(ArgumentError)
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/type_map_by_oid_spec.rb

# Несколько проверок 3/3

```
[1, 0].each do |format|
  it "should convert format #{format} timestamps per TimestampUtcToLocal" do
    regi = PG::BasicTypeRegistry.new
    regi.register_type 0, 'timestamp', nil, PG::TextDecoder::TimestampUtcToLocal
    regi.register_type 1, 'timestamp', nil, PG::BinaryDecoder::TimestampUtcToLocal
    @conn.type_map_for_results = PG::BasicTypeMapForResults.new(@conn, registry: regi)
    res = @conn.exec_params("SELECT CAST('2013-07-31 23:58:59+02' AS TIMESTAMP WITHOUT TIME ZONE),
        CAST('1913-12-31 23:58:59.1231-03' AS TIMESTAMP WITHOUT TIME ZONE),
        CAST('4714-11-24 23:58:59.1231-03 BC' AS TIMESTAMP WITHOUT TIME ZONE),
        CAST('294276-12-31 23:58:59.1231-03' AS TIMESTAMP WITHOUT TIME ZONE),
        CAST('infinity' AS TIMESTAMP WITHOUT TIME ZONE),
        CAST('-infinity' AS TIMESTAMP WITHOUT TIME ZONE)", [], format)

    expect(res.getvalue(0, 0).iso8601(3)).to eq(Time.utc(2013, 7, 31, 23, 58, 59).getlocal.iso8601(3))
    expect(res.getvalue(0, 1).iso8601(3)).to eq(Time.utc(1913, 12, 31, 23, 58, 59.1231).getlocal.iso8601(3))
    expect(res.getvalue(0, 2).iso8601(3)).to eq(Time.utc(-4713, 11, 24, 23, 58, 59.1231).getlocal.iso8601(3))
    expect(res.getvalue(0, 3).iso8601(3)).to eq(Time.utc(294_276, 12, 31, 23, 58, 59.1231).getlocal.iso8601(3))
    expect(res.getvalue(0, 4)).to eq('infinity')
    expect(res.getvalue(0, 5)).to eq('-infinity')
  end
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/basic_type_map_for_results_spec.rb

```ruby
expect(DRb).not_to have_running_server

expect do
  Bisect::Server.run do
    expect(DRb).to have_running_server
    raise 'boom'
  end
end.to raise_error('boom')
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/rspec/core/bisect/server_spec.rb

```ruby
specify '#to_s' do
  channel = Concurrent::Promises::Channel.new
  expect(channel.to_s).to match(/Channel.*unlimited/)
  channel = Concurrent::Promises::Channel.new 2
  expect(channel.to_s).to match(/Channel.*0.*2/)
  channel.push :value
  expect(channel.to_s).to match(/Channel.*1.*2/)
end
```

https://github.com/ruby-concurrency/concurrent-ruby/blob/1982b9/spec/concurrent/edge/channel_spec.rb

# Expectation

```ruby
specify '#(try_)push(_op)' do
  channel = Concurrent::Promises::Channel.new 1

  expect(channel.size).to eq 0
  expect(channel.try_push(:v1)).to be_truthy
  expect(channel.size).to eq 1
  expect(channel.try_push(:v2)).to be_falsey
  expect(channel.size).to eq 1

  channel = Concurrent::Promises::Channel.new 1
  expect(channel.push(:v1)).to eq channel
  expect(channel.size).to eq 1
  thread = in_thread { channel.push :v2 }
  is_sleeping thread
  expect(channel.size).to eq 1
  channel.pop
  expect(channel.size).to eq 1
  expect(thread.value).to eq channel
  channel.pop
  expect(channel.size).to eq 0

  channel = Concurrent::Promises::Channel.new 1
  expect(channel.push(:v1)).to eq channel
  expect(channel.size).to eq 1
  thread = in_thread { channel.push :v2, 0.01 }
  is_sleeping thread
  expect(channel.size).to eq 1
  expect(thread.value).to eq false
  channel.pop
  expect(channel.size).to eq 0
  expect(channel.push(:v3, 0)).to eq true
  expect(channel.size).to eq 1
  thread = in_thread { channel.push :v2, 1 }
  is_sleeping thread
  channel.pop
  expect(channel.size).to eq 1
  expect(thread.value).to eq true

  channel = Concurrent::Promises::Channel.new 1
  expect(channel.push_op(:v1).value!).to eq channel
  expect(channel.size).to eq 1
  push_op = channel.push_op :v2
  expect(channel.size).to eq 1
  expect(push_op.pending?).to be_truthy
  channel.pop
  expect(channel.size).to eq 1
  expect(push_op.value!).to eq channel
  channel.pop
  expect(channel.size).to eq 0
end
```

https://github.com/ruby-concurrency/concurrent-ruby/blob/1982b9/spec/concurrent/edge/channel_spec.rb

# Очевидный комментарий

```
it 'sets :max_length to DEFAULT_MAX_POOL_SIZE' do
  expect(subject.max_length).to eq(
    described_class::DEFAULT_MAX_POOL_SIZE
  )
end
```

concurrent-ruby/blob/1982b9/spec/concurrent/executor/cached_thread_pool_spec.rb

```
specify '#to_s formats as a time', truffle_bug: true do
  expect(subject.to_s).to
    match(/\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2}\.\d{6} \+\d{4} UTC/)
end
```

https://github.com/ruby-concurrency/concurrent-ruby/blob/dadc2a/spec/concurrent/channel/tick_spec.rb

```
it 'renders <video> tag' do
  actual = video_tag('movie.mp4').to_s
  expect(actual).to eq(%(<video src="/assets/movie.mp4"></video>))
end
```

https://github.com/hanami/hanami/blob/a2bdb7/spec/unit/hanami/helpers/assets_helper/video_tag_spec.rb

# Магические циклы

```
it 'should decode integers of different lengths from text format' do
  30.times do |zeros|
    expect(textdec_int.decode('1' + '0' * zeros)).to eq(10**zeros)
    expect(textdec_int.decode(zeros == 0 ? '0' : '9' * zeros)).to eq(10**zeros - 1)
    expect(textdec_int.decode('-1' + '0' * zeros)).to eq(-10**zeros)
    expect(textdec_int.decode(zeros == 0 ? '0' : '-' + '9' * zeros)).to eq(-10**zeros + 1)
  end
  66.times do |bits|
    expect(textdec_int.decode((2**bits).to_s)).to eq(2**bits)
    expect(textdec_int.decode((2**bits - 1).to_s)).to eq(2**bits - 1)
    expect(textdec_int.decode((-2**bits).to_s)).to eq(-2**bits)
    expect(textdec_int.decode((-2**bits + 1).to_s)).to eq(-2**bits + 1)
  end
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/type_spec.rb

# Faker

# TcpGateSwitcher 1/2

```
# This is a transparent TCP proxy for testing blocking behaviour in a time insensitive way.
#
# It works as a gate between the client and the server, which is enabled or disabled by the spec.
# Data transfer can be blocked per API.
# The TCP communication in a C extension can be verified in a (mostly) timing insensitive way.
# If a call does IO but doesn't handle non-blocking state, the test will block and can be caught by an external timeout.
#
#
#   PG.connect                    TcpGateSwitcher
#    port:5444        .-------------------------------------.
#        .--start/stop---------------> T                     |         DB
#  .-----|-----.      |                | /                   |      .------.
#  |    non-   |      |                |/                    |      |Server|
#  |  blocking |      | TCPServer      /           TCPSocket |      | port |
#  |   specs   |------->port 5444-----/   ---------port 5432------->| 5432 |
#  '-----------'      '-------------------------------------'      '------'
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/helpers/tcp_gate_switcher.rb

# TcpGateSwitcher 2/2

```
run_with_gate(200) do |conn, gate|
  conn.setnonblocking(true)

  gate.stop # ТУТ
  data = 'x' * 1000 * 1000 * 30
  res = conn.send_query_params('SELECT LENGTH($1)',
    [data])
  expect(res).to be_nil

  res = conn.flush
  expect(res).to be_falsey

  gate.start # И ЗДЕСЬ
  IO.select(
    nil, [conn.socket_io], [conn.socket_io], 10
  ) until conn.flush
  expect(conn.flush).to be_truthy

  res = conn.get_last_result
  expect(res.values).to eq([[data.length.to_s]])
end
```

https://github.com/ged/ruby-pg/blob/2218eb/spec/pg/connection_spec.rb

# Helpers

```
context 'when all inputs are correct' do
  it 'imports a repository' do
    expect_next_instance_of(Gitlab::GithubImport::SequentialImporter) do |importer|
      expect(importer).to receive(:execute)
    end

    expect_next_instance_of(Project) do |project|
      expect(project).to receive(:after_import)
    end

    import_task
  end
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/tasks/import_rake_spec.rb

```ruby
def expect_next_instance_of(klass, *new_args, &blk)
  stub_new(expect(klass), nil, false, *new_args, &blk)
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/support/helpers/next_instance_of.rb

```ruby
def stub_new(target, number, ordered = false, *new_args, &blk)
  receive_new = receive(:new)
  receive_new.ordered if ordered
  receive_new.with(*new_args) if new_args.present?

  if number.is_a?(Range)
    receive_new.at_least(number.begin).times if number.begin
    receive_new.at_most(number.end).times if number.end
  elsif number
    receive_new.exactly(number).times
  end

  target.to receive_new.and_wrap_original do |*original_args, **original_kwargs|
    method, *original_args = original_args
    method.call(*original_args, **original_kwargs).tap(&blk)
  end
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/support/helpers/next_instance_of.rb

# `without_env_vars` 1/2

```
def without_env_vars(*vars)
  original = ENV.to_hash
  vars.each { |k| ENV.delete(k) }

  begin
    yield
  ensure
    ENV.replace(original)
  end
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/spec_helper.rb

# `without_env_vars` 2/2

```
with_env_vars 'XDG_CONFIG_HOME' => '~/.custom-config' do
  options = parse_options
  expect(options[:formatters]).to eq([['overridden_xdg']])
end

without_env_vars 'XDG_CONFIG_HOME' do
  options = parse_options
  expect(options[:formatters]).to eq([['default_xdg']])
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/rspec/core/configuration_options_spec.rb

# Выразительность

```ruby
FactoryBot.define do
  factory :user, aliases: [:author, :assignee, :recipient, :owner, :resource_owner] do
    email { generate(:email) }
    name { generate(:name) }
    # ..
  end
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/factories/users.rb

```ruby
def textdec_timestamptz_decode_should_fail(str)
  expect(textdec_timestamptz.decode(str)).to eq(str)
end

it 'fails when the timestamp is an empty string' do
  textdec_timestamptz_decode_should_fail('')
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/type_spec.rb

# Детали реализации

```
[
  ['--failure-exit-code', '3', :failure_exit_code, 3],
  ['--pattern', 'foo/bar', :pattern, 'foo/bar'],
  ['--failure-exit-code', '37', :failure_exit_code, 37],
  ['--default-path', 'behavior', :default_path, 'behavior'],
  ['--order', 'rand', :order, 'rand'],
  ['--seed', '37', :order, 'rand:37'],
  ['--drb-port', '37', :drb_port, 37]
].each do |cli_option, cli_value, config_key, config_value|
  it "forces #{config_key}" do
    opts = config_options_object(cli_option, cli_value)
    expect(config).to receive(:force) do |pair|
      expect(pair.keys.first).to eq(config_key)
      expect(pair.values.first).to eq(config_value)
    end
    opts.configure(config)
  end
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/rspec/core/configuration_options_spec.rb

# Метод с ожиданием

```
def expect_parsing_to_fail_mentioning_source(source, options = [])
  expect do
    parse_options(*options)
  end.to raise_error(SystemExit).and output(a_string_including(
                                              "invalid option: --foo_bar (defined in #{source})",
                                              'Please use --help for a listing of valid options'
                                            )).to_stderr
end

context 'defined in $XDG_CONFIG_HOME/rspec/options' do
  it 'mentions the file name in the error so users know where to look for it' do
    file_name = File.expand_path('~/.config/rspec/options')
    create_fixture_file(file_name, '--foo_bar')
    expect_parsing_to_fail_mentioning_source(file_name)
  end
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/rspec/core/configuration_options_spec.rb

# Acceptance

```ruby
specify 'types from a provided types module can be used as setting constructors to coerce values' do
  with_tmp_directory(Dir.mktmpdir) do
    write 'config/app.rb', <<~RUBY
      require "hanami"

      module TestApp
        class App < Hanami::App
        end
      end
    RUBY

    write 'config/settings.rb', <<~RUBY
      module TestApp
        class Settings < Hanami::Settings
          Bool = Types::Params::Bool

          setting :numeric, constructor: Types::Params::Integer
          setting :flag, constructor: Bool
        end
      end
    RUBY

    ENV['NUMERIC'] = '42'
    ENV['FLAG'] = 'true'

    require 'hanami/prepare'

    expect(Hanami.app['settings'].numeric).to eq 42
    expect(Hanami.app['settings'].flag).to be true
  end
end
```

https://github.com/hanami/hanami/blob/a2bdb7/spec/integration/settings/using_types_spec.rb

```ruby
context 'when used in a multithreaded environment with a cassette', with_monkey_patches: :excon do
  it 'properly stubs threaded requests' do
    VCR.use_cassette('/foo') do
      threads = 50.times.map do
        Thread.start do
          Excon.get "http://localhost:#{VCR::SinatraApp.port}/foo"
        end
      end
      threads.each(&:join)
    end

    expect(
      recorded_content_for('foo')
    ).to include('FOO!')
  end
end
```

https://github.com/vcr/vcr/blob/8f5636/spec/acceptance/concurrency_spec.rb

```ruby
def passing_example(fail_if_no_examples)
  "
    RSpec.configure { |c| c.fail_if_no_examples = #{fail_if_no_examples} }

    RSpec.describe 'something' do
      it 'succeeds' do
        true
      end
    end
  "
end

it 'succeeds if fail_if_no_examples set to true' do
  write_file 'spec/example_spec.rb', passing_example(true)
  run_command ''
  expect(last_cmd_stdout).to include('1 example, 0 failures')
  expect(last_cmd_exit_status).to eq(0)
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/integration/fail_if_no_examples_spec.rb

# Библиотеки

```ruby
require "warning"

Warning.ignore(%r{rspec/core})
Warning.ignore(%r{rspec/mocks})
Warning.ignore(/codacy/)
Warning[:experimental] = false if Warning.respond_to?(:[])
```

https://github.com/dry-rb/dry-monads/blob/704c1b/spec/support/warnings.rb

# TestProf 1/2

```
RSpec.describe Projects::BranchesByModeService, feature_category: :source_code_management do
  let_it_be(:user) { create(:user) }
  let_it_be(:project) { create(:project, :repository) }

  let(:finder) { described_class.new(project, params) }
  let(:params) { { mode: 'all' } }
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/services/projects/branches_by_mode_service_spec.rb

# TestProf 2/2

gem 'test-prof', '~> 1.2.3'

https://github.com/test-prof/test-prof/blob/2c8ff6/docs/recipes/let_it_be.md#caveats

# rspec-parameterized 1/3

```
where(:dot_com, :request_user_agent, :qa_user_agent, :result) do
  false | 'qa_user_agent' | 'qa_user_agent' | false
  true  | nil             | 'qa_user_agent' | false
  true  | ''              | 'qa_user_agent' | false
  true  | 'qa_user_agent' | ''              | false
  true  | 'qa_user_agent' | nil             | false
  true  | 'qa_user_agent' | 'qa_user_agent' | true
end
```

https://github.com/gitlabhq/gitlabhq/blob/d1ad38/spec/lib/gitlab/qa_spec.rb

# rspec-parameterized 2/3

```
with_them do
  before do
    allow(Gitlab).to receive(:com?).and_return(dot_com)
    stub_env('GITLAB_QA_USER_AGENT', qa_user_agent)
  end

  subject { described_class.request?(request_user_agent) }

  it { is_expected.to eq(result) }
end
```

https://github.com/gitlabhq/gitlabhq/blob/d1ad38/spec/lib/gitlab/qa_spec.rb

# rspec-parameterized 3/3

```
describe "Hash arguments" do
  where(a: [1, 3], b: [5, 7, 9], c: [2, 4])

  with_them do
    it "sums is even" do
      expect(a + b + c).to be_even
    end
  end
end
```

tomykaira/rspec-parameterized

# Итоги

1. тестируй
2. тестируй важное
3. читай чужой код

# Links

1. https://youtu.be/oNIAJtWuHKg "RSpec. Поддерживаемость"
2. https://bit.ly/3QSGIxh Другие выступления

# Спасибо!